

# Navid Safahieh

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Nationality: New Zealand / Iran

**On the job market 2026–27**

## RESEARCH INTERESTS

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Transportation Economics, Search and Matching, Causal Inference

## EDUCATION

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**Ph.D. in Economics** Expected 2027

Purdue University, Mitch Daniels School of Business, West Lafayette, IN

**M.Sc. in Economics** 2021

University of Tehran, Tehran, Iran

**B.Sc. in Civil and Environmental Engineering** 2018

Sharif University of Technology, Tehran, Iran

**Dissertation Committee:** Cathy Zhang (Chair), Mario J. Crucini, Seunghoon Na, Chong Xiang

## JOB MARKET PAPER

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### Matching Ships to Cargo: Directed Search in Maritime Freight Markets

I develop a directed search model of carrier–customer matching in maritime freight, where carriers post terms and customers direct their search across them. Departing from standard formulations, customers draw a random preparation time upon arrival before they can be matched — the institutional counterpart being laytime and documentation delays before cargo can load. This feature delivers uniqueness of the symmetric equilibrium. I solve the social planner’s problem and the decentralized equilibrium analytically and characterize the relationship between them, and use simulations to trace how equilibrium outcomes respond to changes in market tightness and preparation-time dispersion.

## PUBLICATIONS

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### Modeling the Accuracy of Traffic Crash Prediction Models

- *IATSS Research*, 2022. With Mohammad Hesam Rashidi, Soheil Keshavarz, Parham Pazari, and Amir Samimi.
- Models forecasting accuracy across 31 Iranian provinces using macroeconomic variables and road traffic indicators. Crash variability, average historical crash count, employment ratio, GDP per capita, and traffic exposure are the variables that affect forecast accuracy.

## WORKING PAPERS

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### When Outside Options Shape the Labor Market: A Directed Search Perspective

- Studies how differences in outside options across workers affect wages, firm entry, and job allocation. I develop a directed search model with skilled and unskilled workers and high- and low-tech firms, where only skilled workers hold outside options. Stronger outside options raise posted wages for skilled workers but can reduce high-tech firm entry; as fewer high-tech jobs are created, skilled workers’ expected wages

may fall, and unskilled workers are affected indirectly through changes in firm composition and wage competition.

## The Heterogeneous Effects of Exogenous Shocks on Public Transportation Demand

- With Kowsar Yousefi and Amir Samimi.
- A sudden gasoline price shock increased subway usage by roughly 3.9 percent, implying a cross-price elasticity of about 0.044, while subway usage fell by roughly 42.6 percent during the COVID-19 pandemic.

## CONFERENCE PRESENTATIONS

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- NABE Transfer Pricing / Economic Measurement Conference (TEC2026) 2026 (submitted)
- MEA 90th Annual Meetings, Chicago, IL March 2026

## ADDITIONAL TRAINING

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- Estimation of Structural Models with Conditional Choice Probabilities, Notre Dame, IN Summer 2025
- 14th Annual Research Design for Causal Inference, Northwestern University, IL Summer 2025

## TEACHING EXPERIENCE

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|----------------------------------------------------------------------------|---------------------------------|
| <b>ECON 252 Principles of Macroeconomics (Undergraduate)</b>               | Fall 2025                       |
| <i>Instructor — Outstanding Teaching Award</i>                             | Purdue University               |
| <b>ECON 252 Principles of Macroeconomics (Undergraduate)</b>               | Summer 2025                     |
| <i>Instructor — Distinguished Teaching Award</i>                           | Purdue University               |
| <b>Macroeconomics (PhD)</b>                                                | Fall 2023                       |
| <i>Teaching Assistant to Prof. Seunghoon Na</i>                            | Purdue University               |
| <b>Econometrics (Graduate)</b>                                             | 2019 – 2021                     |
| <i>TA to Profs. Kowsar Yousefi, Atiyeh Vahidmanesh, and Alireza Oryoie</i> | University of Tehran            |
| <b>Game Theory (Graduate)</b>                                              | Fall 2019                       |
| <i>Teaching Assistant to Prof. Ghahreman Abdoli</i>                        | University of Tehran            |
| <b>Econometrics in Transportation (Graduate)</b>                           | 2020 – 2021                     |
| <i>Teaching Assistant to Prof. Amir Samimi</i>                             | Sharif University of Technology |

## FELLOWSHIPS AND AWARDS

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- Bilsland Dissertation Fellowship, Purdue University 2026–27
- Outstanding Teaching Award (ECON 252), Purdue University Fall 2025
- Distinguished Teaching Award (ECON 252), Purdue University Summer 2025

## SERVICE

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- Economics Senator, Business Doctoral Students Association, Purdue University

## PROGRAMMING AND DATA SKILLS

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MATLAB, R, Python, STATA,  $\text{\LaTeX}$

Structural modeling, numerical equilibrium computation, simulation-based methods, large-scale administrative and high-frequency vessel-tracking data (IMF PortWatch, ArcGIS/API pipelines).

## LANGUAGES

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Farsi (Native), English (Fluent)

## REFERENCES

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**Cathy Zhang** (Chair)

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Mitch Daniels School of Business

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